Solid Waste Management

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Energy Content

Unit Energy Content on dry basis $\left(\frac{\mathrm{KJ}}{\mathrm{Kg}}\right) = \frac{\mathrm{KJ}}{\mathrm{Kg}}(\mathrm{discarded})\ \mathrm{x}\ \frac{100}{100-\%\ \mathrm{moisture}}$

Unit Energy Content on ash free basis $\left(\frac{\mathrm{KJ}}{\mathrm{Kg}}\right) = \frac{\mathrm{KJ}}{\mathrm{Kg}}(\mathrm{discarded})\ \mathrm{x}\ \frac{100}{100-\%\ \mathrm{ash}-\ \%\ \mathrm{moisture}}$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Estimate the energy content of a solid-waste sample with the composition given here. What is the content on a dry basis and on an ash-free dry basis?

*Based on a 100 kg sample of waste

Components	Percent by mass*	Moisture Content (%)	Energy, KJ/Kg
Food wastes	15	70	4650
Paper	45	6	16750
Cardboard	10	5	16300
Plastics	10	2	32600
Garden trimmings	10	60	6500
Wood	5	20	18600
Tin Cans	5	3	700

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Determine the moisture content of the sample from the following data:
*Based on a 100 kg sample of waste

Components	Percent by mass*	Energy, KJ/Kg	Total Energy (KJ)
Food wastes	15	4650	69,750
Paper	45	16750	753750
Cardboard	10	16300	163000
Plastics	10	32600	326000
Garden trimmings	10	6500	65000
Wood	5	18600	93000
Tin Cans	5	700	3500
			1474000

Unit Energy Content (as discarded)

$$= \frac{1474000\ KJ}{100\ \text{kg}}$$

$$= 14740\ \text{KJ/Kg}$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Components	Percent by mass*	Moisture Content (%)	Dry mass (kg)
Food wastes	15	70	4.5
Paper	45	6	42.3
Cardboard	10	5	9.5
Plastics	10	2	9.8
Garden trimmings	10	60	4.0
Wood	5	20	4.0
Tin Cans	5	3	4.9
			79.0

$$Moisture\ Content = \frac{100 - 79}{100} x100 = 21\%$$

Unit Energy Content on dry weight basis $\left(\frac{\mathrm{KJ}}{\mathrm{Kg}}\right)$

$$= 1474000 \text{ x} \frac{100}{100 - 21}$$
$$= 18658$$

Unit Energy Content on ash-free dry basis $\left(\frac{\mathrm{KJ}}{\mathrm{Kg}}\right)$

$$= 1474000 \text{ x} \frac{100}{100 - 5 - 21}$$
$$= 19919$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Chemical Content- Ultimate Analysis

- Ultimate Analysis: Percent of Carbon (C), Hydrogen (H), Nitrogen (N), Sulfur (S), Oxygen (O)
- If energy values of the waste as well as its constituents are not available, approximate values maybe determined by using **Modified Dulong Formula**:

$$\frac{\mathbf{KJ}}{\mathbf{Kg}} = \mathbf{337C} + \mathbf{1428}\left(\boldsymbol{H} - \frac{\boldsymbol{O}}{\mathbf{8}}\right) + \mathbf{9}\boldsymbol{S}$$

Where,

C - Percent of Carbon, H - Percent of Hydrogen,

S- Percent of Sulfur, O- Percent of Oxygen

Municipal Solid Waste: A Typical Ultimate Analysis

Element	Range (%dry weight)
Carbon	25-30
Hydrogen	2.5-6.0
Oxygen	15-30
Nitrogen	0.25-1.2
Sulphur	0.02-0.12
Ash	12-30

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. T.V. Ramachandran, MSW NPTEL Course.

Numerical

Derive an approximate chemical formula for the following solid waste sample. Also compute the energy content of the waste:

*Based on a 100 kg sample of waste

Components	Wet mass (%)	Percent by mass (dry weight)					
		Carbon	Hydrogen	Oxygen	Nitrogen	Sulphur	Ash
Food wastes	15	48	6.4	37.6	2.6	0.4	5.0
Paper	45	43.5	6.0	44	0.3	0.2	6.0
Cardboard	10	44	5.9	44.6	0.3	0.2	5.0
Plastics	10	60.0	7.2	22.8	-	-	10.0
Garden trimmings	10	47.8	6.0	38.0	3.4	0.3	4.5
Wood	5	49.5	6.0	42.7	0.2	0.1	1.5

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Components	Wet mass (kg)	Dry mass (kg)	Percent by mass (dry weight)					
			Carbon	Hydrogen	Oxygen	Nitrogen	Sulphur	Ash
Food wastes	15	4.5	2.16	0.29	1.69	0.12	0.02	0.23
Paper	45	42.3	18.40	2.56	18.61	0.13	0.08	2.54
Cardboard	10	9.5	4.18	0.56	4.24	0.03	0.02	0.48
Plastics	10	9.8	5.88	0.71	2.23	-	-	0.98
Garden trimmings	10	4.0	1.91	0.24	1.53	0.14	0.01	0.18
Wood	5	4.0	1.98	0.24	1.71	0.01	-	0.06
	95.0	74.1	34.51	4.58	30	0.43	0.13	4.47

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Components	Wet mass (kg)
Moisture	95-74.1=20.9
Carbon	34.51
Hydrogen	4.58
Oxygen	30.0
Nitrogen	0.43
Sulphur	0.13
Ash	4.47

Convert the moisture content into Hydrogen and Oxygen

Hydrogen = $\frac{2}{18} x 20.9 = 2.32\ kg$

Oxygen = $\frac{16}{18} x 20.9 = 18.58\ kg$

Components	Wet mass (kg)	Percent by mass
Moisture	95-74.1=20.9	34.51
Carbon	34.51	36.3
Hydrogen	4.58+2.32=6.90	7.3
Oxygen	30.0+18.58=48.58	51.1
Nitrogen	0.43	0.5
Sulphur	0.13	0.1
Ash	4.47	4.7
	95.02	100

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Numerical

Components	Wet mass (kg)	Kg/mol	Moles	Sulfur=1.0	Nitrogen=1.0
Carbon	34.51	12.01	2.873	718.2	92.7
Hydrogen	6.90	1.01	6.832	1708	220.4
Oxygen	48.58	16.00	3.036	759	97.9
Nitrogen	0.43	14.01	0.031	7.8	1.0
Sulphur	0.13	32.06	0.004	1.0	0.1

Chemical formula: $C_{92.7}H_{220.4}O_{97.9}N$

$$\boldsymbol{ENERGY\ CONTENT = 337C + 1428\left(H - \frac{O}{8}\right) + 9S = 13546\,\frac{KJ}{Kg}}$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Sources of waste generation

- The overall objective of solid-waste management is to minimize the adverse environmental effects caused by the indiscriminate disposal of solid wastes, especially of hazardous wastes.
- To assess the management possibilities it is important to consider:

(1) materials flow in society,

(2) reduction in raw materials usage,

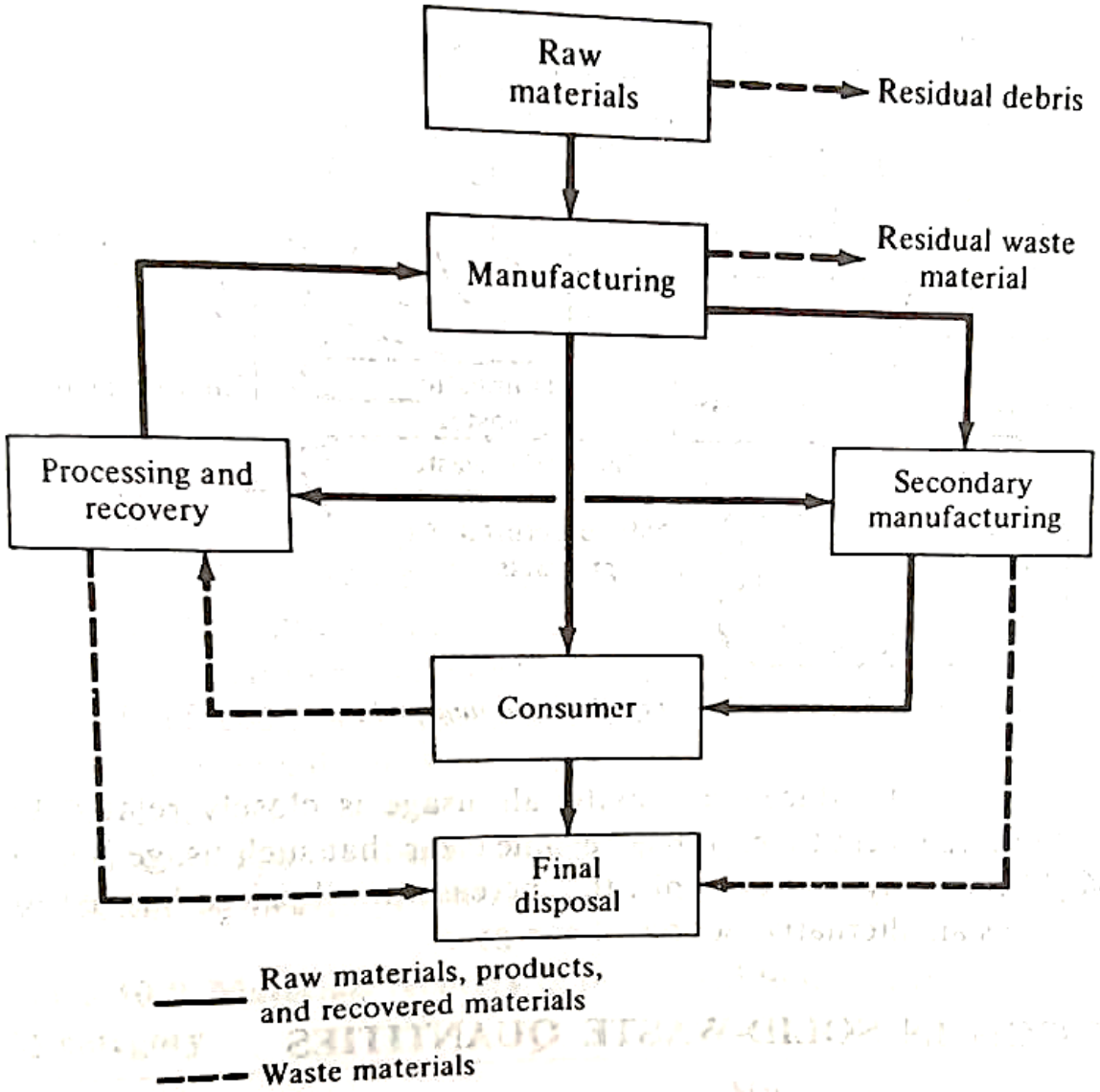
(3) reduction in solid-waste quantities,

(4) reuse of materials,

(5) materials recovery,

(6) energy recovery, and

(7) day-to-day solid-waste management.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

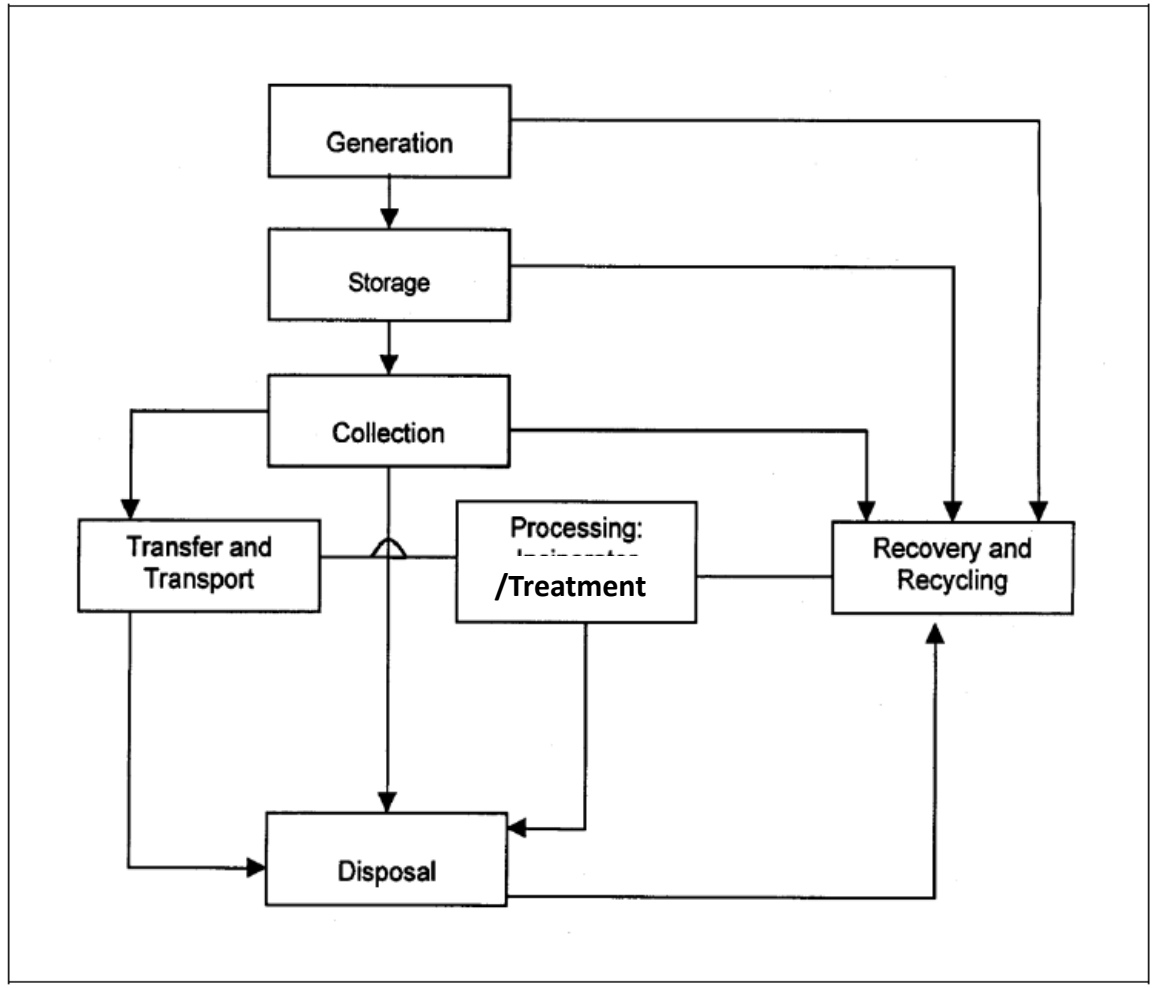
Waste Management Hierarchy

Most preferred- Reduction of waste at source

Least Preferred: Disposal of waste

Source: 1. Environment Protection Agency, Waste management hierarchy, https://www.epa.gov/smm/sustainable-materials-management-non-hazardous-materials-and-waste-management-hierarchy

Functional elements of Solid waste Management



Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Functional elements of Solid waste Management

Functional element	Description
Waste generation	Those activities in which materials are identified as no longer being of value and are either thrown away or gathered together for disposal
On-site handling, storage, and processing	Those activities associated with the handling, storage, and processing of solid wastes at or near the point of generation
Collection	Those activities associated with the gathering of solid wastes and the hauling of wastes after collection to the location where the collection vehicle is emptied
Transfer and transport	Those activities associated with (1) the transfer of wastes from the smaller collection vehicle to the larger transport equipment and (2) the subsequent transport of the wastes, usually over long distance, to the disposal site
Processing and recovery	Those techniques, equipment, and facilities used both to improve the efficiency of the other functional elements and to recover usable materials, conversion products, or energy from solid wastes
Disposal	Those activities associated with ultimate disposal of solid wastes, including those wastes collected and transported directly to a landfill site, semisolid wastes (sludge) from wastewater treatment plants, incinerator residue, compost, or other substances from the various solid-waste processing plants that are of no further use

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Thank you

For any questions, please feel free to reach out to me.